PERIODIC TABLE 周期表

GROUP 族

I	II	III	IV	V	VI	VII	0
							2 **He** 4.0
3 **Li** 6.9	4 **Be** 9.0	5 **B** 10.8	6 **C** 12.0	7 **N** 14.0	8 **O** 16.0	9 **F** 19.0	10 **Ne** 20.2
11 **Na** 23.0	12 **Mg** 24.3	13 **Al** 27.0	14 **Si** 28.1	15 **P** 31.0	16 **S** 32.1	17 **Cl** 35.5	18 **Ar** 40.0
19 **K** 39.1	20 **Ca** 40.1	21 **Sc** 45.0	22 **Ti** 47.9	23 **V** 50.9	24 **Cr** 52.0	25 **Mn** 54.9	26 **Fe** 55.8	27 **Co** 58.9	28 **Ni** 58.7	29 **Cu** 63.5	30 **Zn** 65.4	31 **Ga** 69.7	32 **Ge** 72.6	33 **As** 74.9	34 **Se** 79.0	35 **Br** 79.9	36 **Kr** 83.8
37 **Rb** 85.5	38 **Sr** 87.6	39 **Y** 88.9	40 **Zr** 91.2	41 **Nb** 92.9	42 **Mo** 95.9	43 **Tc** (98)	44 **Ru** 101.1	45 **Rh** 102.9	46 **Pd** 106.4	47 **Ag** 107.9	48 **Cd** 112.4	49 **In** 114.8	50 **Sn** 118.7	51 **Sb** 121.8	52 **Te** 127.6	53 **I** 126.9	54 **Xe** 131.3
55 **Cs** 132.9	56 **Ba** 137.3	57 * **La** 138.9	72 **Hf** 178.5	73 **Ta** 180.9	74 **W** 183.9	75 **Re** 186.2	76 **Os** 190.2	77 **Ir** 192.2	78 **Pt** 195.1	79 **Au** 197.0	80 **Hg** 200.6	81 **Tl** 204.4	82 **Pb** 207.2	83 **Bi** 209.0	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
87 **Fr** (223)	88 **Ra** (226)	89 ** **Ac** (227)	104 **Rf** (261)	105 **Db** (262)													

*	58 **Ce** 140.1	59 **Pr** 140.9	60 **Nd** 144.2	61 **Pm** (145)	62 **Sm** 150.4	63 **Eu** 152.0	64 **Gd** 157.3	65 **Tb** 158.9	66 **Dy** 162.5	67 **Ho** 164.9	68 **Er** 167.3	69 **Tm** 168.9	70 **Yb** 173.0	71 **Lu** 175.0
**	90 **Th** 232.0	91 **Pa** (231)	92 **U** 238.0	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (260)

1 **H** 1.0

atomic number 原子序

relative atomic mass 相對原子質量

Marking Schemes

This document was prepared for markers' reference. It should not be regarded as a set of model answers. Candidates and teachers who were not involved in the marking process are advised to interpret its content with care.

Chemistry
Paper 1

SECTION A

Question No. Part I	Key		Question No. Part II	Key	
1.	C	(79%)	25.	D	(57%)
2.	D	(83%)	26.	C	(86%)
3.	D	(54%)	27.	B	(79%)
4.	C	(52%)	28.	C	(58%)
5.	D	(57%)	29.	B	(69%)
6.	B	(67%)	30.	C	(49%)
7.	A	(74%)	31.	D	(64%)
8.	A	(70%)	32.	A	(62%)
9.	C	(89%)	33.	B	(47%)
10.	B	(74%)	34.	C	(79%)
11.	D	(77%)	35.	A	(56%)
12.	A	(84%)	36.	B	(81%)
13.	A	(74%)			
14.	C	(82%)			
15.	B	(64%)			
16.	C	(76%)			
17.	B	(77%)			
18.	D	(68%)			
19.	A	(66%)			
20.	D	(51%)			
21.	D	(81%)			
22.	A	(73%)			
23.	B	(75%)			
24.	B	(53%)			

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

General Marking Instructions

1. In order to maintain a uniform standard in marking, markers should adhere to the marking scheme agreed at the markers' meeting.

2. The marking scheme may not exhaust all possible answers for each question. Markers should exercise their professional discretion and judgment in accepting alternative answers that are not in the marking scheme but are correct and well-reasoned.

3. In questions asking for a specified number of answers, if a candidate gives more than the required number, the extra answers should not be marked. For instance, in a question asking candidates to provide two examples, if a candidate gives three answers, only the first two answers should be marked.

4. In cases where a candidate answers more questions than required, the answers to all questions should be marked. However, the excess answer(s) receiving the lowest score(s) will be disregarded in the calculation of the final mark.

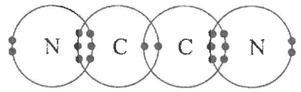
5. Award zero marks for answers which are contradictory.

6. Chemical equations should be balanced except those in reaction schemes for organic synthesis. For energetics, the chemical equations given should include the correct state symbols of the chemical species involved.

7. In the question paper, questions which assess candidates' communication skills are marked with an asterisk (*). For these questions, the mark for effective communication (1 mark per question) will be awarded if the answer is easily understandable. No effective communication marks will be awarded if an answer contains a lot of irrelevant material and/or wrong chemistry concepts.

SECTION B

Part I

			Marks
1.	(a) (i)	The mass number of a ^{13}C atom is greater than that of a ^{12}C atom. / The mass number of a ^{12}C atom is 12 while the mass number of a ^{13}C atom is 13. / A ^{13}C atom has one more neutron than a ^{12}C atom. / A ^{12}C atom has 6 neutrons while a ^{13}C atom has 7 neutrons.	1
	(ii)	Let x be the percentage abundance of ^{12}C. $12x + 13(1 - x) = 12.011$ $x = 0.989 = 98.9\%$	2
	(b)	Similarity: Both diamond and graphite have a giant covalent structure.	1
		Difference: In diamond, each carbon atom forms 4 covalent bonds with other carbon atoms. In graphite, each carbon atom forms 3 covalent bonds with other carbon atoms.	1
		As there are no delocalised electrons in diamond, it is a poor conductor of electricity. As there are delocalised electrons in graphite, it is a good conductor of electricity.	1
	(c)		1

2.	(a)	in the presence of air / oxygen and water	1
	(b)	phenolphthalein	1
	(c)	• Iron loses electrons / is oxidised to form iron(II) ions / $Fe^{2+}(aq)$ ions.	1
		• $Fe^{2+}(aq)$ ions react with potassium hexacyanoferrate(III) to form a blue colour.	1
	(d)	**Y** > iron > **X**	1
		No blue colour is observed / No rusting occurs in set-up 2, which implies that **Y** loses electrons / is oxidised more readily than iron / **Y** gives sacrificial protection for iron.	1
		Blue colour is observed / Rusting occurs in set-up 1, which implies that iron loses electrons / is oxidised more readily than **X**.	

Marks

3. (a) voltmeter / multimeter — 1

(b) To complete the circuit / balance the charges in the two half cells / allow the flow of ions between the two half cells. — 1

(c) (i) $AgNO_3(aq)$ / $Ag^+(aq)$ reacts with $KCl(aq)$ / $Cl^-(aq)$ to form water-insoluble $AgCl(s)$, which will block the ion flow in the salt bridge. — 1

(ii) saturated / concentrated potassium nitrate solution / $KNO_3(aq)$ — 1

(d) (i) Mn — 1
Under using the same metal-metal ion in half-cell **A**, the voltage of chemical cell 3 is the greatest positive value / the highest. — 1

(ii) $+1.15-(-0.46) = +1.61$ — 1

(iii) $Ag^+(aq) + e^- \rightarrow Ag(s)$ — 1

4. (a) (i) No. of moles of H = 3.09 / 1.0 = 3.09 — 2
No. of moles of N =14.42 / 14.0 = 1.03
No. of moles of S = 33.06 / 32.1 = 1.03
No. of moles of O = 49.43 / 16.0 = 3.09

Mole ratio of H : N : S : O = 3 : 1 : 1 : 3
The empirical formula of sulphamic acid is H_3NSO_3.

(ii) Let the molecular formula be $(H_3NSO_3)_n$. — 1
$(3 \times 1.0 + 14.0 + 32.1 + 3 \times 16.0) \times n = 97.1$
$n = 1$
The molecular formula of sulphamic acid is H_3NSO_3.

(b) (i) No. of moles of NaOH that reacted with the $H_2SO_4(aq)$ — 1
$= 2 \times 0.202 \times 24.80 / 1000 = 0.010$

(ii) No. of moles of NaOH that reacted with the sulphamic acid — 1
$= 0.150 \times 100.0 / 1000 - 0.010 = 0.005$

(iii) No. of moles of the sulphamic acid that reacted with NaOH(aq) — 1
$= 0.486 / 97.1 = 0.005$

Mole ratio of H_3NSO_3 : NaOH = 0.005 : 0.005 = 1 : 1

The basicity of sulphamic acid is 1. / Sulphamic acid is a monobasic acid.

(c) $CaCO_3(s) + 2H^+(aq) \rightarrow Ca^{2+}(aq) + CO_2(g) + H_2O(l)$ — 1

Marks

5. (a) Fractional distillation — 1
Different hydrocarbons evaporate and condense as different fractions at different temperatures. — 1

(b) (i) cracking — 1

(ii) Prevent sucking back of water as sudden cooling may break the hot boiling tube. — 1

(iii) The gas mixture contains alkenes which react with $Br_2(aq)$ to give colourless products. — 1

(iv) • Kerosene vaporises quickly without being cracked. — 1
• The broken unglazed porcelain is not hot enough and so cannot function as a catalyst for cracking of kerosene. — 1

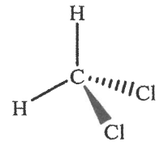
6. (a) (i) $NH_3(aq)$ and $FeSO_4(aq)$ — 1

(ii) iron(II) hydroxide — 1

(b) (i) $Mg(NO_3)_2(aq)$ and $Na_2CO_3(aq)$ — 1

(ii) $Mg^{2+}(aq) + CO_3^{2-}(aq) \rightarrow MgCO_3(s)$ — 1

(c) $H_2SO_4(aq)$ and $Na_2CO_3(aq)$ — 1

7. (a) (i)  — 1

(ii) • C–H bonds are polar because carbon and hydrogen have different electronegativities. — 1
• C–Cl bonds are polar because carbon and chlorine have different electronegativities. — 1

(iii) Dichloromethane is a polar molecule, because the polarities of C–H bonds and C–Cl bonds do not cancel out each other. — 1

(b) • There are weak van der Waals' forces between dichloromethane molecules. — 1
• There are strong hydrogen bonds between ethanol molecules. — 1

Marks

8. (a) Energy released from the reaction = $1.0 \times (50.0 + 50.0) \times 4.20 \times 6.8 \div 1\,000 = 2.856$ kJ — 2
Enthalpy change of neutralisation = $-2.856 \div (1.0 \times 50 \div 1\,000) = -57.12$ kJ mol^{-1}

(b) All acids and alkalis used in experiments 1 to 4 completely dissociate in water. — 1

(c) Enthalpy change for the reaction — 2
$= -50.10 - (-57.12)$
$= +7.02$ kJ mol^{-1}

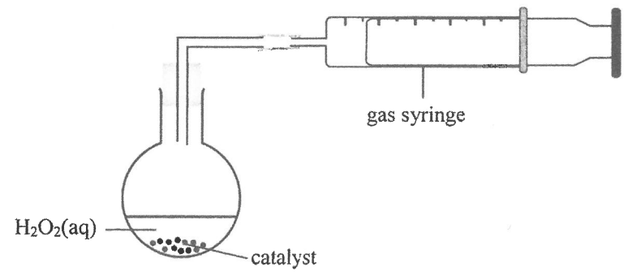
9. Chemical knowledge — 4
- Silver can be obtained by heating silver oxide alone / directly.
- Lead can be obtained by heating a mixture of lead(II) oxide and carbon / charcoal / coke.
- Aluminium can be obtained by electrolysis of aluminium oxide in molten state.
- The more reactive a metal is / The higher the position of a metal in the metal reactivity series is, the more difficult the extraction method of the metal is.

Communication mark — 1

Part II

Marks

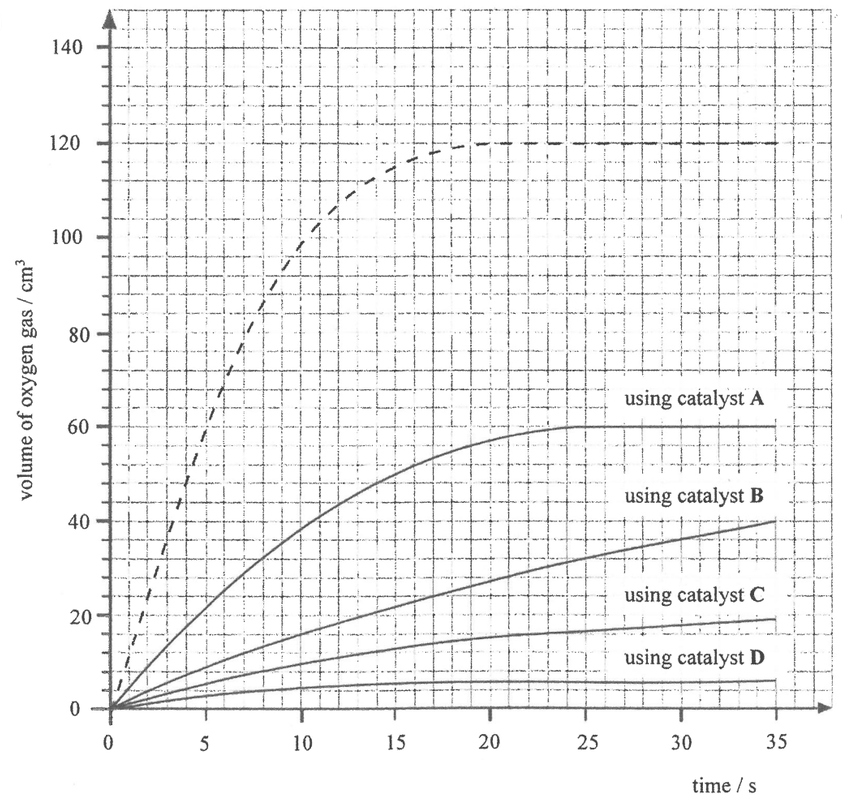
10. (a) — 1



(b) **A**, because it takes the shortest time for the reaction to finish / the initial rate of the reaction is the highest. — 1

(c) 60 cm^3 — 1

(d) — 2

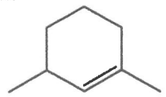


Marks

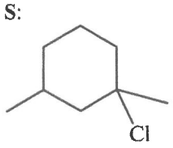
11. (a) 1. $LiAlH_4$, dry ether 2. $H^+(aq)$ — 1
OR
$NaBH_4$

(b) dehydration — 1

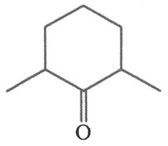
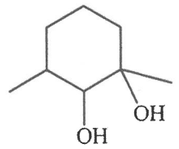
(c) **R**: — 1



(d) **S**: — 1



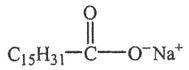
(e) — 1

Q	**R**
	

— 2

12. (a) propane-1,2,3-triol — 1

(b) — 1



(c) (i) There are no two distinct liquid layers after the mixture has rested for a long time. / One persistent milky layer is formed. — 1

(ii) • The ionic head (COO^-) of product **B** dissolves in water / is hydrophilic, while the hydrocarbon tail ($C_{15}H_{31}$) dissolves in oil / is hydrophobic. — 1
• By shaking, the oil breaks up into oil droplets. The repulsion between the negative charges on the oil droplets prevents the oil droplets from joining together again. — 1

Marks

13. (a) • Al_2O_3 is a basic oxide and reacts with HCl(aq) to form $AlCl_3$ and H_2O. / $Al_2O_3(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2O(l)$ — 1

• Al_2O_3 is an acidic oxide and reacts with NaOH(aq) to form $NaAl(OH)_4$ / $NaAlO_2$ and H_2O. / $Al_2O_3(s) + 2NaOH(aq) + 3H_2O(l) \rightarrow 2NaAl(OH)_4(aq)$ — 1

• Aluminium oxide is both an acidic oxide and a basic oxide. — 1

(b) Both S_8 and Cl_2 have a simple molecular structure and there are weak van der Waals' forces between their respective molecules. — 1
The molecular size of S_8 is larger than that of Cl_2 and the van der Waals' forces between S_8 molecules are stronger than those between Cl_2 molecules. — 1

14. Chemical knowledge — 5

Observable change in both cases:
• The equilibrium mixture changes from pink to blue / becomes more intense in blue.

Addition of NaCl(s):
• When the concentration of $Cl^-(aq)$ ions increases, the equilibrium position shifts to the right.
• The equilibrium constant / K_c remains unchanged.

Increase in the temperature:
• As the forward reaction is endothermic, an increase in temperature favours the endothermic side of reaction and the equilibrium position shifts to the right.
• The equilibrium constant / K_c increases.

Communication mark — 1

Paper 2

Marks

1. (a) (i) (1) Haber process — 1

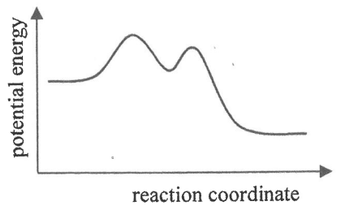
(2) It is a fertiliser. / It can increase the crop yield. — 1

(ii) (1) concentrated sodium chloride solution / brine — 1

(2) No toxic mercury will be produced by membrane electrolytic cells but toxic mercury may leak out from flowing mercury cells. / The operation of membrane electrolytic cells requires less energy than flowing mercury cells. — 1

(iii) Yeast provides enzyme / catalyst. — 1

(b) (i)



— 2

(ii) $\frac{4\times60.0}{2\times58.0+5\times32.0}\times100\% = 87.0\%$ — 1

(iii) $CH_3OH + CO \rightarrow CH_3COOH$ — 1

(iv) HI is regenerated at the end of the reaction. — 1

(v) (1) Toxic CH_3OH / CO and corrosive HI are used in Method 2 while these substances are not used in Method 1. — 1

(2) The atom economy is 100% in Method 2 but not in Method 1. / Method 2 has higher atom economy than Method 1. — 1

(c) (i) Use a gas syringe to follow the change in the volume of the gaseous mixture at regular time intervals. / Use a pressure sensor connected to a data logger to follow the change in the pressure of the gaseous mixture at regular time intervals. — 1

(ii) (1) $a = 1$ — 1

(2) $\log \frac{k2}{k1} = \frac{Ea}{2.3R}\left(\frac{1}{T1} - \frac{1}{T2}\right)$ — 2

$\log\left(\frac{k}{4.29\times10^{-4}}\right) = \frac{108500}{2.3\ (8.31)}\left(\frac{1}{318} - \frac{1}{338}\right)$

$k = 4.88\times10^{-3}\ s^{-1}$

(iii) (1) **B** — 1

(2) • An increase in temperature can increase the average kinetic energy of the molecules. — 1

• There are more molecules having kinetic energy equal to or greater than the activation energy. — 1

• This will increase the effective collision frequency / the number of effective collisions per unit time. — 1

Marks

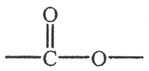
2. (a) (i) • When a rubber band is stretched, the polymer chains move only slightly without breaking the disulphur cross-linkages. — 1

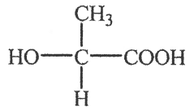
• When the stretching force is removed, the polymer chains are still held together by the cross-linkages, and they return to their original position. — 1

(ii) (1) $NHCOCH_3$ / acetylamino group — 1

(2) There are hydrogen bonds between the polymer chains of chitin. — 1

(iii) smectic phase — 1

(b) (i) ester group /  — 1

(ii)  — 1

(iii) PLA is formed by joining up the monomer molecules repeatedly with the elimination of small molecules. — 1

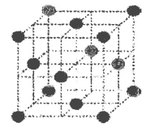
(iv) Corn starch is a renewable resource. — 1

(v) PLA is non-toxic / biocompatible. — 1

(vi) Not suitable, because PLA is a thermoplastic. — 1

(vii) Because ethyl ethanoate molecules and PLA molecules are polar, while hexane molecules are non-polar. / The intermolecular attractions in ethyl ethanoate and PLA are similar while that in hexane is different. — 1

(c) (i) 8 — 1

(ii) (1)  — 1

(2) A-B-C type packing — 1

(iii) The deduction is incorrect. As the temperature increases, the iron metal expands on heating, so the density of iron at 1 450 °C is lower. — 1

(iv) (1) Kevlar

• It has a higher tensile strength than iron. — 1

• It has a lower density than iron. — 1

(2) • There are more hydrogen bonds between the polymer chains of Kevlar than those of nylon-6,6. — 1

• Kevlar has many benzene rings which restrict the movement between the polymer chains, but nylon-6,6 does not have. — 1

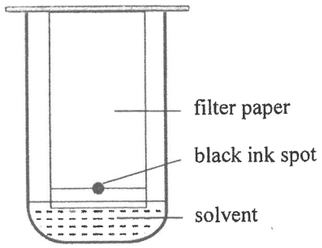
Marks

3. (a) (i) • Add 2,4-dinitrophenylhydrazine. 1
• Compound **A** forms a yellow / orange / red precipitate while compound **B** does not. 1

OR

• Add $Na_2CO_3(aq)$ / $NaHCO_3(aq)$. / Add deionised water and then add Mg(s).
• Compound **B** gives out a colourless gas while compound **A** does not.

(ii) 2



(iii) Any one of the following: 1
• These solvents are immiscible with water.
• These solvents do not react with the organic compounds.
• These solvents can dissolve more organic compounds than water can.
• These solvents have low boiling points / can evaporate easily.

(b) (i) To ensure that all $PbCO_3(s)$ and $CuCO_3(s)$ have completely reacted with dilute $HNO_3(aq)$. 1

(ii) A deep blue solution is formed. 1

(iii) (1) • Washing is to remove water soluble impurities and drying is to remove water. 1
• To obtain a more accurate mass of $Pb(OH)_2(s)$. 1

(2) lead(II) hydroxide / $Pb(OH)_2$ 1

(3) Percentage by mass of $PbCO_3$ in the solid sample 2

$$= \frac{1.47}{(207.2+16.0\times2+1.0\times2)} \times (207.2 + 12.0 + 16.0 \times 3) \div 2.03 \times 100\%$$

$$= 80.22\%$$

(c) (i) The infra-red spectrum of the indoor air sample shows a strong absorption peak in the region from 1 680 cm^{-1} to 1 800 cm^{-1} corresponding to the carbonyl group / C=O bond of formaldehyde. 1

(ii) HCO^+ 1

(iii) (1) colorimeter 1

(2) The absorbance is directly proportional to the concentration of formaldehyde. 1

(3) From the graph, the concentration of formaldehyde in solution **S** is 0.25 mg dm^{-3}. 2
Mass of formaldehyde = 0.25 × 0.03 = 0.0075 mg

(4) Concentration of formaldehyde in the indoor air sample 1
= 0.0075 / 0.1 = 0.075 mg m^{-3} < 0.1 mg m^{-3}
Therefore, the quality of the air inside the room is good.

(5) Instrumental analytical methods are sensitive enough to measure very low levels of formaldehyde. 1

Candidates' Performance

Paper 1

Paper 1 consisted of two sections: Section A (multiple-choice questions) and Section B (conventional questions). Sections A and B each comprised two parts: Part I and Part II. Part I contained questions mainly on Topics I to VIII of the curriculum, while Part II contained questions mainly on Topics IX to XII. All questions in both sections were compulsory.

Section A (multiple-choice questions)

This section consisted of 36 multiple-choice questions. The mean score was 24.8. The performance of candidates was generally good. Some misconceptions were revealed by candidates in their performance on the following items:

1. For Q.4, about half of the candidates correctly chose the answer C. It should be noted that 0.1 M **Y**(aq) is a dilute solution. The pH of an aqueous solution of an acid depends on the concentration of $H^+(aq)$ ions. The concentration of $H^+(aq)$ ions depends on both the concentration and the strength of the acid. If **Y** is a strong monobasic acid, the expected pH of 0.1 M **Y**(aq) should be 1 (pH = –log 0.1). The actual pH (i.e. 2.91) is higher than 1, which implies that **Y** does not completely ionise in water, therefore **Y** should be a weak acid. However, about a quarter of the candidates wrongly thought that **Y** is a strong acid.

Q.4 **Y** is a monobasic acid. The pH of 0.1 M **Y**(aq) is 2.91. Which of the following statements concerning the 0.1 M **Y**(aq) is correct ?

A.	It is a concentrated solution of a weak acid.	(11%)
B.	It is a concentrated solution of a strong acid.	(9%)
C.*	It is a dilute solution of a weak acid.	(52%)
D.	It is a dilute solution of a strong acid.	(28%)

2. For Q.24, most candidates were able to decide that the second statement is a true statement. Some candidates did not recognise that dilute nitric acid can act as a strong oxidising agent and have redox reaction with copper. About a quarter of the candidates wrongly thought that the first statement is false and chose option C as the answer.

Q.24 Consider the following statements and choose the best answer :

1st statement	**2nd statement**
Copper can react with dilute nitric acid but not with dilute ethanoic acid.	Nitric acid is a strong acid while ethanoic acid is a weak acid.

A.	Both statements are true and the 2nd statement is a correct explanation of the 1st statement.	(18%)
B.*	Both statements are true and the 2nd statement is NOT a correct explanation of the 1st statement.	(53%)
C.	The 1st statement is false but the 2nd statement is true.	(25%)
D.	Both statements are false.	(4%)

3. For Q.30, about half of the candidates were able to calculate the correct numerical value of the equilibrium constant K_c. Some candidates neglected to notice that $H_2O(l)$ is in the same physical state as the other three compounds in the equilibrium system, and they did not consider the equilibrium concentration of $H_2O(l)$ in the calculation of K_c. About a quarter of the candidates wrongly chose option D as the answer. They might think that as the total volume of the reaction mixture was not known, the equilibrium concentrations of the four compounds in the reaction mixture could not be found and K_c could not be determined. Candidates should note that the term, total volume of the reaction mixture (V), will be cancelled in the following expression when calculating K_c.

$$K_c = \frac{\left[\frac{\text{number of moles of } CH_3COOC_2H_5(l)}{V}\right]_{eqm}\left[\frac{\text{number of moles of } H_2O(l)}{V}\right]_{eqm}}{\left[\frac{\text{number of moles of } CH_3COOH(l)}{V}\right]_{eqm}\left[\frac{\text{number of moles of } C_2H_5OH(l)}{V}\right]_{eqm}}$$

Q.30 Refer to the following catalysed reaction at a certain temperature :

$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$

A mixture of 2.50 mol of $CH_3COOH(l)$ and 2.50 mol of $C_2H_5OH(l)$ was placed in a closed container and kept at this temperature. When equilibrium was attained, 0.85 mol of $CH_3COOH(l)$ remained in the reaction mixture. What is the numerical value of the equilibrium constant K_c for the above reaction at this temperature ?

A. 0.27 (11%)
B. 2.28 (11%)
C.* 3.77 (49%)
D. K_c cannot be determined because the total volume of the mixture is not known. (29%)

Section B (conventional questions)

Question Number	Performance in General
1	The performance of candidates was good. In part (a)(i), a very high percentage of the candidates were able to give one difference between a ^{12}C atom and a ^{13}C atom, such as 'having different numbers of neutrons or different masses'. A very small number of candidates wrongly stated that a ^{12}C atom and a ^{13}C atom have different numbers of protons or electrons. In part (a)(ii), about three quarters of the candidates were able to correctly calculate the percentage abundance of ^{13}C. A small number of candidates gave an incorrect expression of the percentage in their answers. In part (b), a small number of candidates were able to give a correct and complete answer. About two thirds of the candidates correctly stated that both diamond and graphite have giant covalent structures. However, a small number of candidates wrongly stated that graphite has a simple molecular structure. A high proportion of candidates were able to state that the carbon atoms in diamond and graphite are bonded by strong covalent bonds. However, only a small number of candidates explicitly mentioned that each carbon atom forms four covalent bonds with other carbon atoms in diamond, while each carbon atom only forms three covalent bonds with other carbon atoms in graphite. About two thirds of the candidates were able to show that only graphite has delocalised electrons which make it a good conductor of electricity. Some candidates failed to state 'delocalised electrons' but instead used less precise terms, such as 'mobile electrons' or 'free electrons'. A few candidates wrongly stated that covalent bonds form between diamond molecules or graphite molecules. In part (c), about half of the candidates were able to correctly draw the electron diagram for a C_2N_2 molecule. A small number of candidates neglected the lone pair electrons on the nitrogen atoms.
2	The performance of candidates was good. In part (a), a very high percentage of candidates were able to state that the presence of air / oxygen and water is essential for the rusting of iron. In part (b), about three quarters of the candidates were able to state that phenolphthalein changed to a pink colour in the two set-ups. A small number of candidates gave hydroxide ions as the answer without mentioning phenolphthalein. In part (c), about a quarter of the candidates were able to give a complete answer. Some candidates wrongly stated that iron(III) ions, instead of iron(II) ions, react with potassium hexacyanoferrate(III) and turn it blue. Some candidates merely stated that iron(II) ions were formed, without explaining the formation of iron(II) ions due to iron losing electrons or being oxidised. Some candidates only mentioned the appearance of the blue colour due to the presence of iron(II) ions, without clearly stating which chemical species turned blue. In part (d), about half of the candidates were able to arrange the three metals in the correct decreasing order of reactivity, but less than half of them were able to give a correct explanation. Many candidates failed to make their explanations based on the experimental observations. Some candidates only mentioned the relative reactivity of the metals in their answers, without clearly referring to the ability of the metals to lose electrons or carry out sacrificial protection.

Question Number	Performance in General
3	The performance of candidates was good. In part (a), a very high percentage of candidates were able to correctly identify the apparatus **W** as a voltmeter or a multimeter. Some candidates misspelt the word 'voltmeter' as 'volmeter' or 'voltameter'. In part (b), about three quarters of the candidates were able to suggest one correct function of a salt bridge. However, a small number of candidates wrongly stated that a salt bridge allows electrons to pass through or they provided vague answers such as 'It provides mobile ions.'. In part (c)(i), just under half of the candidates were able to correctly explain why KCl(aq) cannot be used. They stated that KCl(aq) reacts with $AgNO_3$(aq) to form insoluble AgCl(s), which blocks the ion flow in the salt bridge. Some candidates only mentioned that KCl(aq) would react with the electrolytes used in the chemical cell without explicitly mentioning the formation of AgCl(s). Moreover, some candidates wrongly considered Cl^-(aq) to be a reducing agent in the experiment and that Cl_2(g) would be formed. In part (c)(ii), about two thirds of the candidates were able to suggest a suitable solution that can be used for preparing the salt bridge. Some candidates wrongly suggested using an acid (e.g. HCl(aq)) or an alkali (e.g. NaOH(aq)) for preparing the salt bridge. A small number of candidates failed to recognise the problem that arose from using KCl(aq) and wrongly suggested using KBr(aq) or NaCl(aq) to make the salt bridge. In part (d)(i), about two thirds of the candidates were able to correctly state and explain that manganese is the strongest reducing agent among the four metals. A small number of candidates wrongly stated 'Mn^{2+}' or 'Mn/Mn^{2+}' as the strongest reducing agent. Moreover, some candidates only mentioned Mn as being the most reactive metal in their answers without referring to the cell voltage shown in the table. In part (d)(ii), less than half of the candidates were able to calculate and give the correct voltage for the chemical cell 4. Many candidates only wrote the numerical value without including the '+' sign. In part (d)(iii), about two thirds of the candidates were able to write the correct half equation. A small number of candidates confused the anode with the cathode and wrongly wrote the half equation for the oxidation of Mn to Mn^{2+}.
4	The performance of candidates was good. In part (a)(i), a very high percentage of candidates were able to calculate and state the correct empirical formula of sulphamic acid. A small number of candidates only wrote the empirical formula without showing their calculations. In part (a)(ii), about three quarters of the candidates were able to deduce the correct molecular formula of sulphamic acid. A small number of candidates only stated 'n=1' without writing the molecular formula, or only wrote the molecular formula without providing any deduction. In part (b)(i), about two thirds of the candidates were able to calculate the correct number of moles of NaOH that reacted with H_2SO_4. A small number of candidates failed to realise that the mole ratio between NaOH and H_2SO_4 in the reaction is 2:1, resulting in their incorrect answers. In part (b)(ii), about half of the candidates were able to perform the calculation correctly. In part (b)(iii), about half of the candidates were able to deduce the correct basicity of sulphamic acid. Some candidates wrongly deduced that sulphamic acid is a tribasic acid by solely counting the number of hydrogen atoms in the molecular formula. In part (c), less than half of the candidates were able to write the correct ionic equation for the reaction between $CaCO_3$(s) and H^+(aq). Some candidates were not able to balance the equation correctly. A small number of candidates wrongly considered that 'Ca' or 'CaO' was formed in the reaction. There were also some candidates who failed to recognise that $CaCO_3$ is an insoluble solid and incorrectly wrote 'CO_3^{2-}' instead of $CaCO_3$(s) as one of the reactants.

Question Number	Performance in General
5	The performance of candidates was satisfactory. In part (a), a very high proportion of the candidates were able to state 'fractional distillation' as the physical method to obtain kerosene from petroleum. Just under half of the candidates were able to explain the underlying principle of fractional distillation. Some candidates wrongly considered that both petroleum and kerosene were single substances, and that the underlying principle of fractional distillation is based on the difference in the boiling points of petroleum and kerosene. In part (b)(i), a very high proportion of the candidates were able to correctly name the chemical process. In part (b)(ii), about half of the candidates were able to suggest a reason why the delivery tube should be taken out of the water before removing the heat source. Some candidates made a mistake in the answer, such as 'To prevent sucking back of water which may crack the delivery tube.'. In part (b)(iii), about two thirds of the candidates were able to point out the presence of alkenes in the collected gas mixture. Some candidates merely stated that bromine was reduced to bromide ions to account for the decolourisation of bromine water. In part (b)(iv), about a quarter of the candidates were able to suggest a correct reason, such as 'kerosene vaporised quickly and passed the broken unglazed porcelain without cracking', to explain why cracking would not occur significantly. Some candidates wrongly stated 'There is no broken unglazed porcelain and so no catalyst is present for the reaction.'. Some candidates even proposed that kerosene would burn to give carbon dioxide and water if only the mineral wool with kerosene is heated strongly.
6	The performance of candidates was good. In part (a)(i), about two thirds of the candidates were able to give the correct answer. In part (a)(ii), less than half of the candidates were able to give the correct name of the green precipitate. Some candidates overlooked the question and gave the formula $Fe(OH)_2$ as the answer. In part (b)(i), about two thirds of the candidates were able to give the correct answer. In part (b)(ii), about half of the candidates were able to write the correct ionic equation. In part (c), a high proportion of the candidates were able to write the correct answer.
7	The performance of candidates was satisfactory. In part (a)(i), about three quarters of the candidates were able to draw the three-dimensional structure of a dichloromethane molecule correctly. In part (a)(ii), less than half of the candidates were able to explain why the bonds in a dichloromethane molecule are polar based on the difference in electronegativities between the bonding atoms of the covalent bonds in the dichloromethane molecule. In part (a)(iii), less than half of the candidates were able to suggest a reason to explain why a dichloromethane molecule is polar. Some candidates provided wrong answers, such as 'the polar bonds cannot cancel out each other' and 'C-Cl bonds and C-H bonds will cancel each other'. In part (b), nearly half of the candidates were able to state that the intermolecular forces in dichloromethane and ethanol are van der Waals' forces and hydrogen bond respectively, while hydrogen bond is stronger than van der Waals' forces. Some candidates did not compare the relative strength of these intermolecular forces to explain why ethanol has a higher boiling point.

Question Number	Performance in General
8	The performance of candidates was satisfactory. In part (a), about two thirds of the candidates were able to calculate the enthalpy change of neutralisation between HCl(aq) and NaOH(aq) using the given experimental data. Some candidates carelessly gave the answer without the '–' sign. In part (b), just under half of the candidates were able to point out that all the acids and alkalis used in experiments 1 to 4 completely dissociate in water. Some candidates gave incorrect answers like 'the number of moles of water formed in the four experiments are the same' and 'the concentrations of acids and alkalis used in the four experiments are the same'. In part (c), just under half of the candidates were able to calculate the enthalpy change for the reaction. Some candidates carelessly gave the answer without the '+' sign.
9	The performance of candidates was satisfactory. About half of the candidates were able to write different methods to extract silver, lead and aluminium from their oxides. Many candidates were able to demonstrate a complete understanding of the relation between the extraction method of a metal and its position in the metal reactivity series. Some candidates confused 'burn' with 'heat' and they wrongly gave an answer as 'burn lead(II) oxide with carbon to get lead metal'. Some candidates did not present their answers clearly and precisely, such as 'Aluminium can be obtained by electrolysis.'. Some candidates only used chemical equations to show different reactions involving the three metal oxides, without elaborating how these reactions can be used to extract metals. These candidates were not able to demonstrate effectively the communication skills required by this question.
10	The performance of candidates was good. In part (a), about two thirds of the candidates were able to draw a properly labelled diagram showing how to measure the volumes of oxygen gas formed in the experiment. Most of the candidates chose to draw a labelled gas syringe. However, some candidates drew the set-up diagram poorly or misspelt the word 'syringe'. In part (b), about half of the candidates were able to give the correct answer. Although many candidates were able to state that '**A**' is the most effective catalyst, some of them did not give a clear explanation. They failed to explicitly state that the volumes of oxygen gas produced should be compared within the same period of time. In part (c), about half of the candidates were able to correctly state that the theoretical final volume of oxygen gas produced in the experiment using catalyst **B** is 60 cm^3. Many candidates wrongly gave '40 cm^3' as the answer, they failed to recognise that the four experiments should give the same amount of oxygen gas when the reaction completed. In part (d), about half of the candidates were able to give a correct sketch to show the expected result. Some candidates failed to notice that the number of moles of $H_2O_2(aq)$ used was doubled in the fifth experiment, and they wrongly drew a curve that levelled off at 60 cm^3. A small number of candidates did not read the question carefully and ignored the given information 'The reaction completed at the 20th second.'. They drew a curve that reached the final volume at the wrong time.

Question Number	Performance in General
11	The performance of candidates was satisfactory. In part (a), about half of the candidates were able to suggest the correct reagent and condition needed for the conversion of **P** to **Q**. Many candidates were able to recognise that the conversion can be achieved by using a proper reducing agent, such as lithium aluminium hydride or sodium borohydride, while some of them failed to provide the exact condition. Common mistakes included '$LiAlH_4(aq)$', '$LiAlH_4$ in $H^+(aq)$' and '$NaBH_4$ in $H^+(aq)$'. Some candidates incorrectly wrote the chemical formula for lithium aluminium hydride as '$AlLiH_4$'. In part (b), about two thirds of the candidates were able to correctly state the type of reaction involved in the conversion of **Q** to **R** as 'dehydration'. A small number of candidates gave incorrect answers, such as 'hydrolysis', 'condensation' or 'oxidation'. In part (c), about two thirds of the candidates were able to draw the correct chemical structure of **R**. Some candidates, however, carelessly omitted one or both 'CH_3' groups in their answers. In part (d), just under half of the candidates were able to draw the correct chemical structure of the major product **S**. Some candidates failed to apply Markovnikov's rule on the addition reaction between **R** (an alkene) and HCl(g), and they wrongly drew the structure of the minor product. Similar to part (c), a small number of candidates carelessly omitted one or both 'CH_3' groups in their answers. In part (e), about a third of the candidates were able to correctly identify that **Q** and **R** can react with $KMnO_4(aq)$ and draw the accurate structures of the corresponding organic products. About half of the candidates were able to correctly state that **Q** can react with $KMnO_4(aq)$ and draw the correct structure of the ketone product. Some candidates were not able to give the structure of the product formed from the reaction of **R** and $KMnO_4(aq)$.
12	The performance of candidates was fair. In part (a), about a quarter of the candidates were able to write 'propane-1,2-3-triol' as the systematic name of product **A**. Many candidates wrongly wrote the name as 'propan-1,2,3-triol'. A small number of candidates wrongly gave the trivial name 'glycerol'. In part (b), about half of the candidates were able to draw the correct structure of product **B**. Some candidates were confused about products **A** and **B**, and wrongly drew the structure of propane-1,2,3-triol. A small number of candidates wrongly gave the structure as '$C_{15}H_{31}COOH$'. In part (c)(i), about two thirds of the candidates were able to correctly suggest an expected observation when product **B** is added to an oil-water mixture and shaken. A small number of candidates failed to describe the observation clearly and precisely. They gave vague answers such as 'water and oil mixed together'. In part (c)(ii), only a quarter of the candidates were able to give a correct and complete explanation for the formation of emulsion mentioned in part (i). Many candidates gave incomplete or imprecise explanations. They only mentioned that product **B** has a hydrophilic part and a hydrophobic part without clearly indicating which part of the structure of **B** is hydrophilic and which part is hydrophobic. Some candidates only mentioned that there are repulsions between the oil droplets, but they failed to state clearly that the repulsions are between the negative charges on the oil droplets.

Question Number	Performance in General
13	The performance of candidates was satisfactory. In part (a), about a third of the candidates were able to correctly describe the acid-base behavior of aluminium oxide with reference to the respective products formed when aluminium oxide reacts with HCl(aq) and with NaOH(aq). Many candidates correctly stated that $AlCl_3$ is formed from the reaction between Al_2O_3 and HCl. However, a small number of candidates failed to mention that H_2O is also one of the products in the reaction. Some candidates were not able to write the correct chemical formula of the product formed from the reaction between Al_2O_3 and NaOH. About half of the candidates were able to state that Al_2O_3 is a basic oxide as it reacts with HCl, and it is also an acidic oxide when it reacts with NaOH. However, a small number of candidates only mentioned that Al_2O_3 can react with both acids and bases, without further elaborating on the acid-base behaviour of Al_2O_3 with reference to the reactions that occur or products formed. In part (b), about a third of the candidates were able to correctly explain why sulphur is a solid but chlorine is a gas at room conditions. Many candidates were able to recognise that both sulphur and chlorine have a simple molecular structure and their molecules are held by weak van der Waals' forces. However, some candidates wrongly considered sulphur to have a giant covalent structure, and that a large amount of energy is needed to break the strong covalent bonds between sulphur atoms in order to melt sulphur into a liquid. Additionally, many candidates were able to state that the molecular size of sulphur is larger than that of chlorine, but some candidates failed to mention that sulphur exists as S_8 molecules.
14	The performance of candidates was satisfactory. About a quarter of the candidates were able to give a correct, complete and well-presented answer, while about half of the candidates were able to give a partially correct answer. Many candidates were able to recognise that the equilibrium positions will shift to the right in both cases. However, a small number of candidates failed to clearly mention that the shift of equilibrium position is due to the increase in the concentration of $Cl^-(aq)$ ions upon the addition of NaCl(s) or the fact that the forward reaction is endothermic. Some candidates did not precisely describe the expected colour change, they only vaguely stated that the colour of the equilibrium mixture will become 'darker'. Moreover, quite a number of candidates did not read the question carefully and did not mention whether there was any change in the equilibrium constant in their answers. Some candidates only mentioned that the equilibrium constant increases upon increasing the temperature, but they did not explicitly mention that the equilibrium constant remains unchanged when NaCl(s) is added to the mixture at a constant temperature.

Paper 2

Paper 2 consisted of three sections. Section A contained questions set on Topic XIII 'Industrial Chemistry', Section B on Topic XIV 'Materials Chemistry' and Section C on Topic XV 'Analytical Chemistry'. Candidates were required to attempt all questions in two of the sections.

Question Number	Popularity (%)	Performance in General
Section A: 1	98	The performance of candidates in part (a) was fair. In part (a)(i)(1), about two thirds of the candidates were able to name the industrial process to produce ammonia. Some candidates wrongly spelt the process as 'Harbour' process. In part (a)(i)(2), about two thirds of the candidates were aware that ammonium sulphate is a fertiliser which can alleviate the problem of inadequate food supply in the world. In part (a)(ii)(1), about half of the candidates were able to suggest the raw material used in the flowing mercury cells. Some candidates wrongly stated that mercury was the raw material. In part (a)(ii)(2), about a third of the candidates were able to give one advantage of using membrane electrolytic cells over flowing mercury cells. Some candidates did not write the answer in a comparative sense. In part (a)(iii), about a quarter of the candidates were able to state the function of yeast in the fermentation of sugars precisely. The performance of candidates in part (b) was satisfactory. In part (b)(i), about three quarters of the candidates were able to draw an energy profile for the catalysed reaction involved in Method 1 with correct labels of axes. Some candidates wrongly labelled the x-axis as the 'reaction coordination' or 'time'. Some candidates wrongly labelled the y-axis as 'kinetic energy' or 'energy level'. In part (b)(ii), about two thirds of the candidates were able to calculate the atom economy of Method 1. In part (b)(iii), about two thirds of the candidates were able to write the chemical equation for the overall reaction involved in Method 2. Some candidates wrongly included HI as both a reactant and a product in the chemical equation. In part (b)(iv), about a third of the candidates were able to suggest that HI can be considered as a catalyst in Method 2 because HI is regenerated in the process. Some candidates only mentioned that HI is produced in Step (3) without stating that HI is used in Step (1). In part (b)(v)(1), about a third of the candidates were able to suggest a reason why Method 1 can be considered as 'greener' than Method 2. In part (b)(v)(2), just under half of the candidates were able to suggest a reason why Method 2 can be considered as 'greener' than Method 1. The performance of candidates in part (c) was satisfactory. In part (c)(i), a small number of the candidates were able to suggest how the progress of the decomposition reaction can be followed. Some candidates only mentioned the parameter to be followed without mentioning that the parameter concerned should be monitored over a period of time. In part (c)(ii)(1), about half of the candidates were able to write a numerical value for ***a***. Some candidates did not recognise that the numerical value of ***a*** can be obtained with reference to the unit of the given rate constant. In part (c)(ii)(2), just under half of the candidates were able to calculate the rate constant using the given Arrhenius equation. In part (c)(iii)(1), a very high proportion of the candidates were able to state that curve **B** can represent the Maxwell-Boltzmann distribution curve at a temperature higher than T_1. In part (c)(iii)(2), about half of the candidates were able to explain why increasing the temperature will increase the rate of a gaseous reaction.

Question Number	Popularity (%)	Performance in General
Section B: 2	6	The performance of candidates in part (a) was very poor. In part (a)(i), about a quarter of the candidates were able to explain the elasticity of the rubber band in terms of the disulphur cross-linkages between the polymer chains. In part (a)(ii)(1), a small number of the candidates were able to correctly state what **X** is. In part (a)(ii)(2), about a third of the candidates were able to point out that the hardness of the exoskeleton of lobsters is related to the formation of intermolecular hydrogen bonds in chitin. In part (a)(iii), a small number of the candidates were able to write the correct name of the liquid crystal phase. The performance of candidates in part (b) was very poor. In part (b)(i), about a quarter of the candidates were able to state the functional group of PLA. In part (b)(ii), about a third of the candidates were able to draw the correct structure of the monomer of PLA. Some candidates drew incorrect structures like $OHCH(CH_3)COOH$, $O{=}CH(CH_3)COOH$ and $HOCH(CH_3)CHO$. In part (b)(iii), just under half of the candidates were able to point out that there would be elimination of small molecules during the formation of PLA from its monomers. In part (b)(iv), about a quarter of the candidates were able to mention that corn starch is a renewable resource. Some candidates carelessly mentioned that corn starch is a renewable energy source. In part (b)(v), about a third of the candidates suggested that PLA is a biocompatible material and hence it can be used to make surgical threads. In part (b)(vi), about a quarter of the candidates were able to state that PLA is a thermoplastic, while compression moulding is suitable to be used to produce items made of thermosetting plastic. In part (b)(vii), a small number of the candidates were able to point out that both PLA and ethyl ethanoate are polar molecules while hexane is non-polar, therefore the intermolecular attractions in PLA and ethyl ethanoate are similar, but that in hexane is different. Some candidates gave incorrect answers, such as 'There are hydrogen bonds formed between PLA and ethyl ethanoate.'. The performance of candidates in part (c) was very poor. In part (c)(i), about a third of the candidates were able to give the correct coordination number of an iron atom. In part (c)(ii)(1), about a third of the candidates were able to correctly draw a unit cell of a cubic close-packed crystal structure. In part (c)(ii)(2), a small number of candidates were able to state the arrangement of iron atoms in a cubic close-packed crystal structure correctly. In part (c)(iii), a very small number of the candidates were able to point out that the deduction is incorrect as there would be an increase in the volume of iron with an increase in temperature for the same mass of iron. Many candidates gave the incorrect answer: 'At 25 °C and 1450 °C, iron has the same crystal structure and the same arrangement pattern of iron atoms in the unit cell. Therefore, the deduction is correct.'. In part (c)(iv)(1), about half of the candidates were able to suggest that Kevlar is a better choice for making bullet-proof vests with reference to the greater relative tensile strength and the lower density of Kevlar. In part (c)(iv)(2), a very small number of the candidates were able to suggest reasons to explain why Kevlar has a higher rigidity than nylon-6,6. Some candidates made mistakes such as 'Nylon-6,6 cannot form intermolecular hydrogen bonds between the polymer chains while Kevlar can.', 'Kevlar has cross-links between polymer chains but nylon-6,6 does not.' and 'The hydrogen bonds in Kevlar are stronger than those in nylon-6,6.'.

Question Number	Popularity (%)	Performance in General
Section C: 3	96	The performance of candidates in part (a) was fair. In part (a)(i), about half of the candidates were able to suggest a chemical test to distinguish between $CH_3CH_2CH_2COCH_3$ and $CH_3CH_2CH_2COOH$. Some candidates wrongly stated that only $CH_3CH_2CH_2COOH$ would give a colourless gas when both compounds were treated with Na_2CO_3 without mentioning that the presence of water is needed for the reaction to occur. Some candidates did not spell '2,4-dinitrophenylhydrazine' correctly. In part (a)(ii), about half of the candidates were able to draw a labelled diagram for the set-up required for the paper chromatography experiment. Some candidates drew only the chromatogram without recognising that the set-up should include a piece of filter paper and an appropriate solvent in a container. In part (a)(iii), less than half of the candidates were able to suggest one property of the solvents rendering them to be used in the liquid-liquid extraction. The performance of candidates in part (b) was fair. In part (b)(i), less than half of the candidates were able to explain why excess dilute $HNO_3(aq)$ was added in Step (1). Some candidates only mentioned that it is to remove either $PbCO_3$ or $CuCO_3$ but not to remove all the solid carbonates. In part (b)(ii), about a third of the candidates were able to state another expected observation. In part (b)(iii)(1), about half of the candidates were able to explain why it is necessary to wash and dry the precipitate. In part (b)(iii)(2), about half of the candidates were able to state the precipitate as being $Pb(OH)_2$. Some candidates wrongly stated the precipitate as being PbO, $Cu(OH)_2$ or $Cu(NH_3)_2$. In part (b)(iii)(3), just under half of the candidates were able to estimate the percentage by mass of $PbCO_3$ in the sample. The performance of candidates in part (c) was satisfactory. In part (c)(i), about half of the candidates were able to suggest how the presence of the functional group of formaldehyde can be detected by using infra-red spectroscopy. Some candidates wrongly wrote the 'wavenumber' as 'wavelength'. Some candidates did not mention the functional group corresponding to the absorption peak. In part (c)(ii), about half of the candidates were able to suggest one chemical species corresponding to the peak at m/z = 29. Some candidates wrongly wrote the species as $HCO{-}^+$, HCO or $C_2H_5^+$. In part (c)(iii)(1), about two thirds of the candidates were able to name the instrument **Y**. Some candidates gave the wrong spelling of 'colorimeter'. Some candidates wrongly wrote the name of the method 'colorimetry' instead of the name of the instrument. In part (c)(iii)(2), about two thirds of the candidates were able to state the relationship between absorbance and concentration of formaldehyde in the given graph. Some candidates only stated that the absorbance is proportional to the concentration of formaldehyde without mentioning that it is directly proportional. In part (c)(iii)(3), about half of the candidates were able to calculate the mass of formaldehyde in solution **S**. In part (c)(iii)(4), less than half of the candidates were able to explain whether the quality of the air inside the room is good. In part (c)(iii)(5), about two thirds of the candidates were able to suggest why instrumental analytical methods are generally used for measuring the formaldehyde levels in indoor air.

School-based Assessment

All school candidates sitting for HKDSE Chemistry have to participate in School-based Assessment (SBA). In 2024, the Chemistry SBA adopted an interim resumption arrangement. A total of 12 329 students from 445 schools submitted their SBA marks. With the experience acquired over the past years, the implementation was generally smooth in most of the participating schools.

Based on the assessment data and samples of students' work submitted by participating schools, students' performance was in general satisfactory and within the expectations of the assessment requirements. To address potential discrepancies in the marking standard among individual teachers and schools, moderation on the SBA scores submitted by schools was performed according to both statistical methods and professional judgement. It was reported that 51.7% of schools fell into the 'within the expected range' category, while 31.5% of schools had marks higher than expected, and 16.8% of schools had marks lower than expected this year. It was observed that the majority of schools with deviations only differed slightly from the expected range. This implies that most of the teachers have a good understanding of SBA implementation, and hence the marking standards are generally appropriate.

To ensure that teachers have a good understanding of the requirements and the principles of the assessment methods of the SBA, a Chemistry SBA conference and group meetings are held every year. The conference and group meetings can provide teachers with general comments on and a summary of the SBA implementation, and the latest updates of the SBA requirements and administrative operations. Teachers can make use of the resources and support introduced at the conference to integrate practical work into lessons. Furthermore, the Education Bureau and the Hong Kong Examinations and Assessment Authority also provide training courses and resources for new teachers, to enhance their knowledge and skill as well as to build up confidence in implementing SBA in schools.

To provide continuing support for teachers and to ensure fair implementation of the SBA, two supervisors and 24 district coordinators were appointed to address enquiries from teachers about SBA implementation and ensure that schools were administering the SBA in accordance with the stipulated guidelines. Phone calls, email correspondences, district group meetings and school visits were conducted to establish close connections between the district coordinators and the teachers. The said communication channels among the supervisors, district coordinators and teachers can enhance mutual understanding of the SBA implementation in schools. Based on the feedback and suggestions obtained from various sources, teachers and students have a better understanding of the essence and the requirements of the SBA. Nonetheless, some comments and recommendations are given below so that further improvement to the implementation of SBA can be made:

1. Variety of Experiments
   Even though there is no strict stipulated requirement regarding the types of experiments selected for SBA tasks, besides including volumetric analysis and qualitative analysis in the task list, it is definitely beneficial to students' learning if they are exposed to a wider variety of experiment types. It is noted that a very small number of schools did not include a 'titration' exercise in their volumetric analysis experiments, which may not be beneficial to the development of the holistic laboratory-based learning experience. Experiments from different topics like 'Chemistry reactions and energy', 'Rate of reaction', 'Chemical equilibrium', 'Organic Synthesis' and 'Analytical chemistry' can be used as SBA tasks. Conducting these types of experiments can enrich students' practical experience as well as strengthen students' practical skills and analytical thinking skills. For qualitative analysis, detection of ions (cations and / or anions) is required, but not the detection of organic species. Moreover, 'Other experiments' refers to experiments suggested in the Curriculum and Assessment Guide, but not volumetric analysis nor detection of ions. It is encouraging to observe that some schools, with sufficient lesson time and other resources, arranged 'preparative experiments' for their students.

2. Variety of Written Work
   Practical worksheets, experiment-related quizzes and brief / detailed laboratory reports, etc. are all acceptable formats of written work. Teachers generally designed these tasks in a professional manner. Moreover, it is encouraging that most students can follow the instructions given by teachers in accomplishing the written work. Although there are no stipulated requirements in the SBA guidelines regarding the types of written work, writing detailed laboratory reports is definitely important for students studying science. Organising and writing a laboratory report in the correct format as well as presenting the data and experimental findings properly are very important skills.

3. Assessment Criteria and Mark Scale
   To carry out a valid assessment of students' attainment in practical tasks in SBA, teachers can make use of the most suitable assessment criteria for different types of practical tasks and review them from time to time. In addition, a systematic mark scale or assessment rubrics can be deployed to increase the reliability of the assessment.

4. Use of feedback to promote learning
   Providing feedback to students, in particular with written remarks and marks through submitted reports, is important for facilitating student learning. Students can be encouraged to review their own work and avoid making the same type of mistakes in the future. They can also discuss with their teachers their performance when carrying out experiments and in related written tasks, so as to reinforce their laboratory learning experiences. Teachers can provide students with oral, written and other forms of feedback to promote learning through the SBA.

5. Students' performance in recording and analysing the data obtained from experiments
   It was observed that students frequently made mistakes in recording the experimental data and performing calculations. These mistakes include using incorrect significant figures in data recording and calculations, using incorrect units for numerical data, and carrying out the calculations incorrectly. It was also observed that quite a number of students incorrectly recorded the burette readings in titrations using numbers with only one decimal place. In addition, students showed little attention to the handling of graphs and drawing set-up diagrams, which are essential expected learning outcomes. Students should pay more attention to these areas.

6. Prevention of plagiarism
   Students should complete the assessment tasks honestly and responsibly in accordance with the stipulated requirements. They will be subject to severe penalties for proven malpractice, such as plagiarising others' work. The HKDSE Examination Regulations stipulate that a candidate may be liable to disqualification from the subject concerned or the whole of the examination, or suffer a mark or grade penalty for breaching the regulations. Students can refer to the information leaflet HKDSE Examination - Information on School-based Assessment (http://www.hkeaa.edu.hk/DocLibrary/Media/Leaflets/SBA_pamphlet_E_web.pdf) for guidance on how to properly acknowledge sources of information quoted in their work.

Conclusion

For the implementation of SBA in the 2024 HKDSE, students' performance is generally satisfactory, and teachers have expressed a smooth running of the SBA in their lessons. With the experience acquired in the previous cohorts, most teachers have a clear understanding about the requirements and expected goals of SBA. Teachers are also experienced in selecting appropriate practical tasks and assessing the abilities of their students. Finally, it is observed that most of the queries and challenges that teachers and students encountered have been handled appropriately.